

Midterm 1 Part a

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I. QUESTIONS (54 POINTS)

1. In which of the following machine learning applications there is a supervisor and it tries to estimate a discrete entity from a set of possibilities?

- a) Association
- b) Classification
- c) Regression
- d) Unsupervised Learning
- e) Reinforcement Learning

2. In which of the following machine learning applications the output of the system is a set of actions and the important thing is the policy?

- a) Association
- b) Classification
- c) Regression
- d) Unsupervised Learning
- e) Reinforcement Learning

Given the following classification dataset $\mathcal{X}_1$. Answer questions 3-7 accordingly.

$$\mathcal{X}_1 = \begin{bmatrix} & x_1 & x_2 & x_3 & C \\ x^{(1)} & 4 & 2 & green & yes \\ x^{(2)} & 6 & 3 & yellow & no \\ x^{(3)} & 7 & 2 & blue & yes \\ x^{(4)} & 3 & 1 & green & no \\ x^{(5)} & 3 & 1 & blue & no \end{bmatrix}$$

3. How many classes are there in this dataset?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

4. How many features are discrete?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

5. How many features are continuous ?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

6. What is the mean vector of features 1 and 2 ?

- a) [4.6 1.8]
- b) [1.8 4.6]
- c) [6 2]
- d) [4 1]
- e) [4.4 1.6]

$$\mathcal{X}_1 = \begin{bmatrix} & x_1 & x_2 & x_3 & C \\ x^{(1)} & 4 & 2 & green & yes \\ x^{(2)} & 6 & 3 & yellow & no \\ x^{(3)} & 7 & 2 & blue & yes \\ x^{(4)} & 3 & 1 & ? & no \\ x^{(5)} & 3 & 1 & blue & no \end{bmatrix}$$

7. If feature 3 of $x^{(4)}$ is missing, what will be the imputation value for it?

- a) green
- b) yellow
- c) blue
- d) gray
- e) None

8. If you have d discrete features where each feature has $f_1, f_2, \dots, f_d$ distinct values, what will be the number of continuous features in the converted dataset, when all discrete features are converted to continuous using 1-of-L encoding?

- a) d
- b) dL
- c) $f_1 + f_2 + \dots + f_d$
- d) $f_1 * f_2 * \dots * f_d$
- e) $d!$

Given the dataset in the questions above, the classification decisions for two classifiers are given below. Answer questions 9-11 accordingly.

$Classifier_1$	$Classifier_2$
no	yes
no	no
yes	yes
yes	no
yes	yes

9. What is the accuracy of $Classifier_1$?

- a) 0
- b) 0.2
- c) 0.8
- d) 0.4
- e) 1

10. If yes is the positive class, then what is TP (True positive) of $Classifier_1$?

- a) 0
- b) 1
- c) 2
- d) 3
- e) 4

11. If yes is the positive class, then what is the precision of $Classifier_1$?

- a) 1 / 1
- b) 1 / 2
- c) 1 / 3
- d) 3 / 4
- e) 2 / 3

Given the following regression dataset $\mathcal{X}_2$. Answer question 12-13 accordingly.

$$\mathcal{X}_2 = \begin{bmatrix} & x_1 & x_2 & r \\ x^{(1)} & 4 & 2 & 4.2 \\ x^{(2)} & 6 & 3 & 3.6 \\ x^{(3)} & 7 & 2 & 2.4 \\ x^{(4)} & 3 & 1 & 6.2 \end{bmatrix}$$

12. What is the average regression value of the dataset?

- a) 3.8
- b) 4.0
- c) 4.1
- d) 4.3
- e) 4.5

13. How many instances are there in this dataset ?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

Given the dataset in the questions above, the regression decisions for two regressors are given below. Answer questions 20-21 accordingly.

$Regressor_1$	$Regressor_2$
4.0	4.0
6.0	3.6
3.0	2.2
7.0	6.4

14. What is the mean absolute error of $Regressor_1$?

- a) 0.7
- b) 0.8
- c) 0.9
- d) 1.0
- e) 1.1

15. If we have a K class classification dataset, how many one-vs-one 2 class problems can be defined for this dataset?

- a) $K - 1$
- b) K
- c) $K + 1$
- d) $(K - 1)K/2$
- e) K^2

16. If we have a K class classification dataset, how many one-vs-rest 2 class problems can be defined for this dataset?

- a) $K - 1$
- b) K
- c) $K + 1$
- d) $(K - 1)K/2$
- e) K^2

17. What is the square root of variance?

- a) mean
- b) variance
- c) gauss
- d) standard deviation

18. When the covariance of two variables are nonzero and their variances are different, what is the shape of the isoprobability contour plot?

- a) square
- b) ellipse
- c) circle
- d) ellipse with some degree inclined

19. Which of the following classifiers is linear?

- a) Qda
- b) Only Lda
- c) Only Lda and Nearest Mean
- d) Lda, Nearest Mean, Naive Bayes

20. In which classifier the covariance matrices of all classes are different?
- Qda
 - Lda
 - Nearest Mean
 - Naive Bayes
21. In which classifier the covariance matrices of all classes are same, but the covariance matrix is diagonal and the items in the diagonal are the same?
- Qda
 - Lda
 - Nearest Mean
 - Naive Bayes
22. Which of the following datasets is used to tune the hyperparameters of the classifier?
- Train set
 - Test set
 - Validation Set
23. What will be the training test percentage if we do K-fold cross-validation?
- $(K-1)/K$ percent
 - $100 * (K-1)/K$ percent
 - $1/K$ percent
 - $100/K$ percent
 - 100K percent
24. What will be the testing test percentage if we do K-fold cross-validation?
- $(K-1)/K$ percent
 - $100 * (K-1)/K$ percent
 - $1/K$ percent
 - $100/K$ percent
 - 100K percent
25. If you have N instances in the dataset, what will be the size of the bootstrap sample?
- $N/10$
 - $10N$
 - $2N$
 - N
 - N^2
26. What is the maximum value of area under curve for a classifier?
- 1
 - 2
 - 10
 - 0.9
 - e
27. What is the value of area under curve for dummy classifier?
- 1
 - 2
 - 0.5
 - 0.9
 - e
28. Which of the following experiment design techniques has the lowest space complexity?
- Best guess
 - One factor at a time
 - Factorial design
29. Which of the following algorithms has the lowest space complexity?
- Qda
 - Lda
 - Nearest Mean
 - Naive Bayes
30. Which of the following algorithms has the highest space complexity?
- Qda
 - Lda
 - Nearest Mean
 - Naive Bayes
31. No ... theorem that says there is no such thing as the best learning algorithm?
- Late Breakfast
 - Free Lunch
 - Spoon
 - Last Supper
32. If the classifier is trained less than it was needed and it didn't learn enough, then the classifier is ...?
- Overfitting
 - Underfitting
 - Fitting
33. If the classifier has large training and test errors, then the classifier is ...?
- Overfitting
 - Underfitting
 - Fitting
34. Which of the following a subset selection technique starts with empty subset of features and adds features one by one?
- Forward selection
 - Backward selection
 - Floating selection
 - Exhaustive search

35. Which of the following a subset selection technique starts with full set of features and removes features one by one?

- a) Forward selection
- b) Backward selection
- c) Floating selection
- d) Exhaustive search

36. What is the number of possible subsets of d variables?

- a) d
- b) d^2
- c) 2^d
- d) d^3

37. Which of the following is a feature extraction technique?

- a) Forward selection
- b) Backward selection
- c) Floating selection
- d) PCA

38. If the eigenvalues of the covariance matrix are $\lambda_1 = 2$, $\lambda_2 = 1$, $\lambda_3 = 0.5$, $\lambda_4 = 0.1$, and $\lambda_5 = 0$; then with proportion of variance explained 90 percent, what will be the reduced dimensionality?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

39. If the eigenvalues of the covariance matrix are $\lambda_1 = 2$, $\lambda_2 = 1$, $\lambda_3 = 0.5$, $\lambda_4 = 0.1$, and $\lambda_5 = 0$; then with proportion of variance explained 99 percent, what will be the reduced dimensionality?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

Assume you wish to use the K-nearest neighbors algorithm on this dataset.

A	B	C	D	Output
1	1	1	1	F
0	0	1	0	T
0	1	0	1	F
1	1	0	0	F
1	0	1	1	F
0	0	0	0	?

40. Which class will be assigned to the test instance (last instance) if $k = 1$?

- a) T
- b) F

41. Which class will be assigned to the test instance (last instance) if $k = 3$?

- a) T
- b) F

42. If 95 percent of the unit distribution lies between $(-1.96, 1.96)$, how many percent of the data lies between $(-\infty, 1.96)$?

- a) 2.5
- b) 5
- c) 50
- d) 95
- e) 97.5

43. If 90 percent of the unit distribution lies between $(-1.64, 1.64)$, how many percent of the data lies between $(-1.64, 0)$?

- a) 2.5
- b) 5
- c) 45
- d) 95
- e) 99

44. Which of the following is a two-sided null hypothesis resulting in a two-sided test?

- a) $H_0 : \mu_0 = \mu_1$
- b) $H_0 : \mu_0 \leq \mu_1$
- c) $H_0 : \mu_0 \geq \mu_1$

45. Which of the following is a statistical test to compare two classification algorithms on a single validation set?

- a) K-Fold Cv t test
- b) McNemar's test
- c) 5x2 cv Paired t test
- d) 5x2 cv Paired F test